

CS5691 Assignment 4

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Semester: Jan-May 2022

Report

1.1 K-Nearest Neighbours algorithm

In this part, we built a classifier based on the k-nearest neighbours algorithm for all the given four datasets. We analyzed the performance of our model by applying dimensionality reduction through techniques like PCA, LDA.

1.1.1 Synthetic Dataset

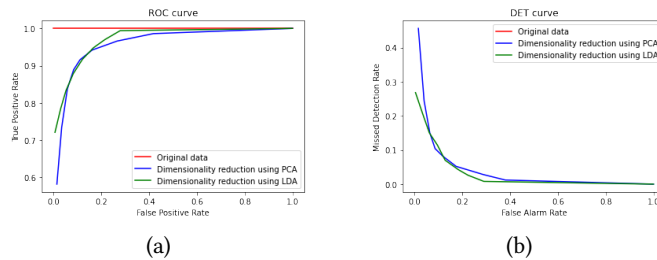


FIGURE 1.1: ROC and DET curves for synthetic data ($K = 8$)

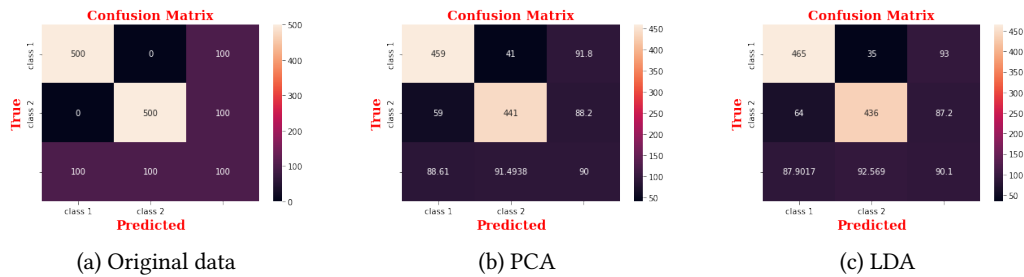


FIGURE 1.2: Confusion matrices for synthetic data ($K = 8$)

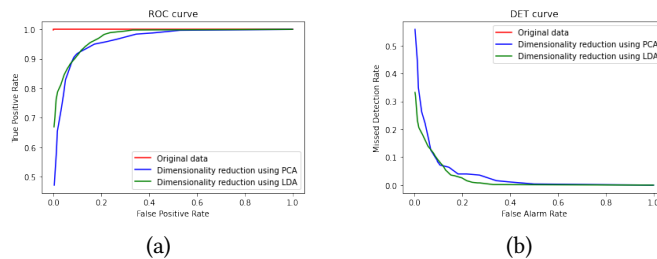


FIGURE 1.3: ROC and DET curves for synthetic data ($K = 20$)

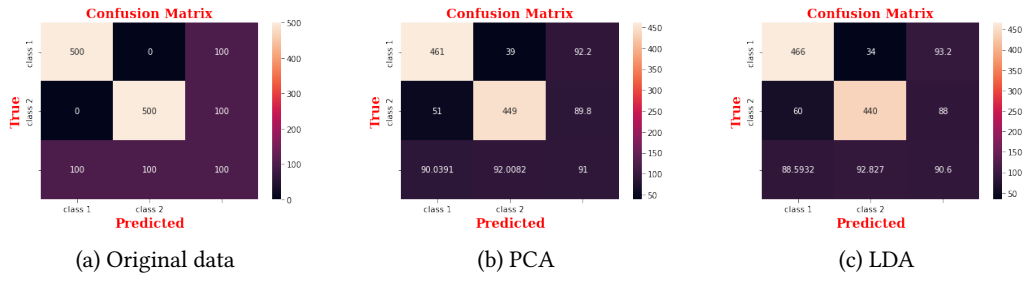


FIGURE 1.4: Confusion matrices for synthetic data(K = 20)

1.1.2 Effect of dimensionality reduction

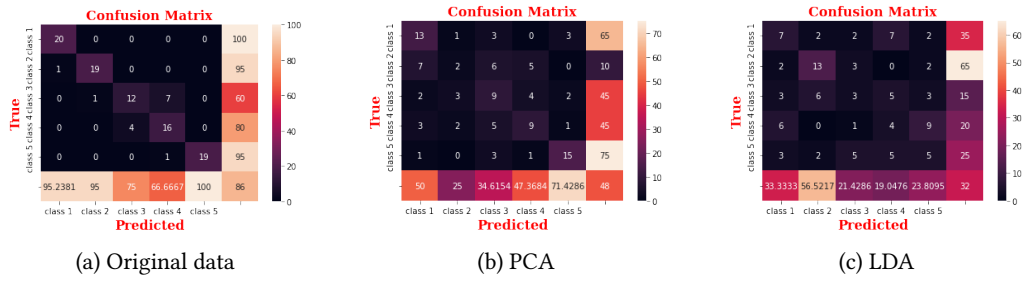


FIGURE 1.5: New dimension = 10

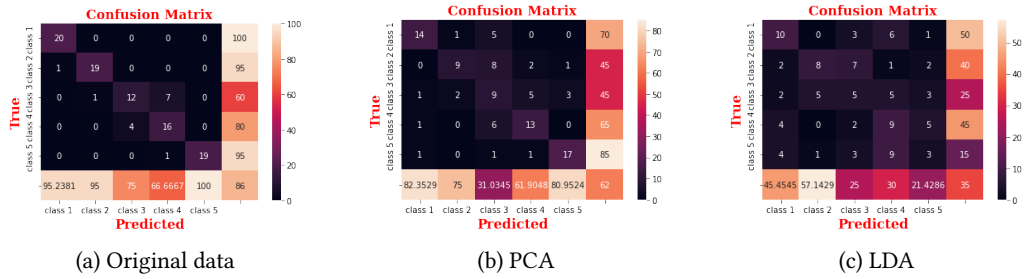


FIGURE 1.6: New dimension = 100

1.2 Logistic Regression

Results are arranged in the order of without dimension reduction,PCA,LDA. For PCA and LDA results are for fixed value of reduced dimension (NX4). N is largest data time-step. We used Scipy.signal Resample to make equal length time series.

1.2.1 Synthetic Dataset

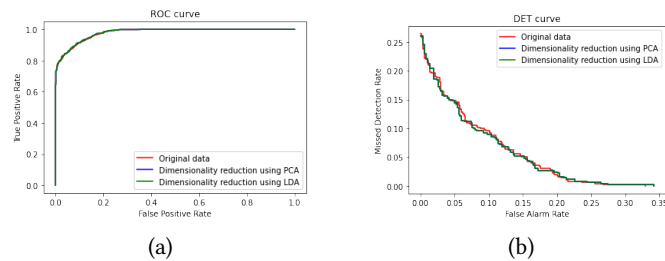


FIGURE 1.7

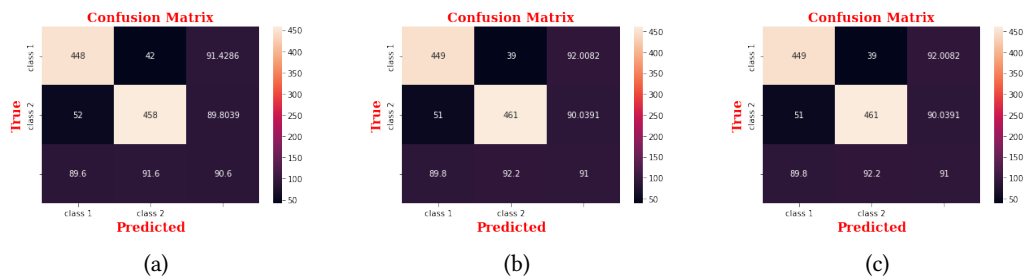


FIGURE 1.8

1.2.2 Image Dataset

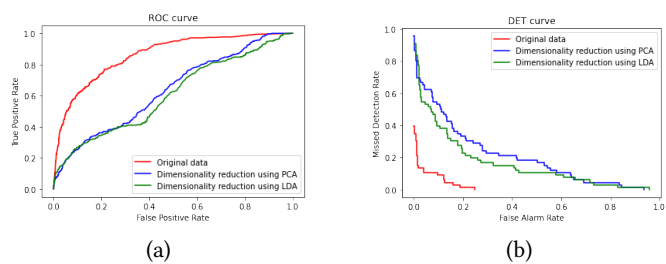


FIGURE 1.9

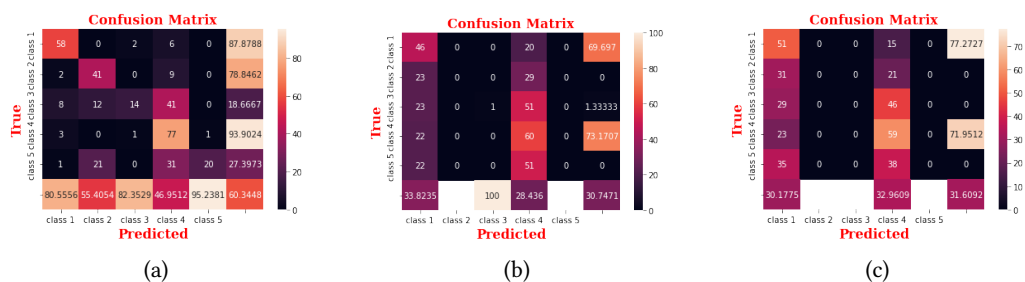


FIGURE 1.10

1.2.3 Isolated digits Dataset

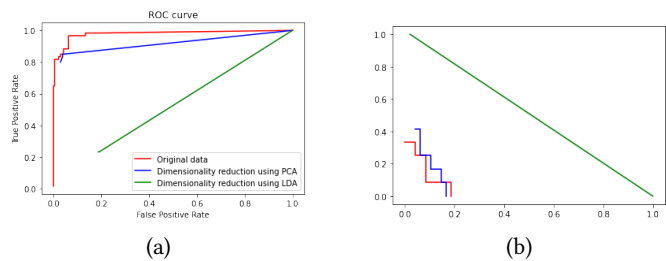


FIGURE 1.11

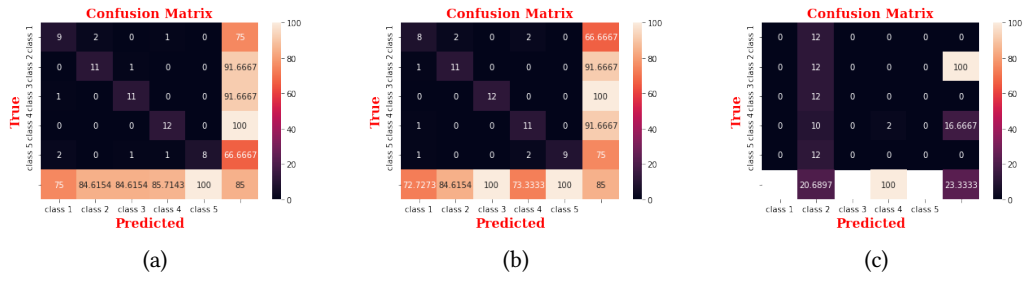


FIGURE 1.12

1.2.4 Handwritten character Dataset

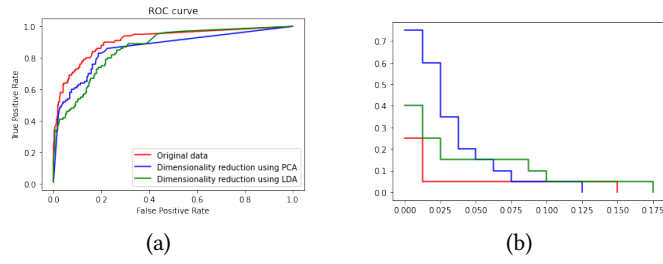


FIGURE 1.13

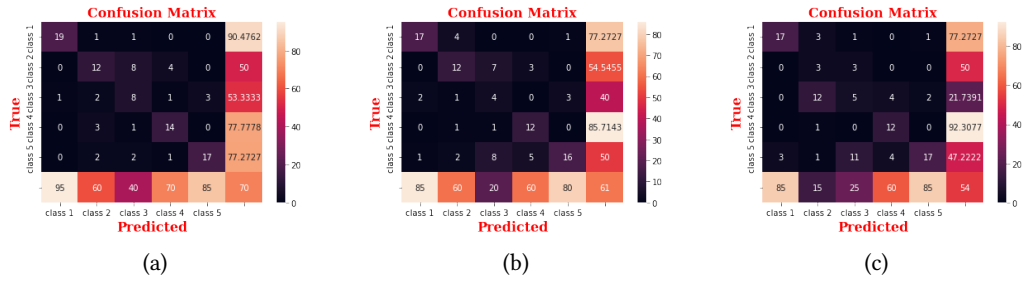


FIGURE 1.14

1.3 Support Vector Machines

Results are arranged in the order of without dimension reduction, PCA, LDA. For PCA and LDA results are for fixed value of reduced dimension (NX4). N is largest data time-step. We used Scipy.signal Resample to make equal length time series.

1.3.1 Synthetic Dataset

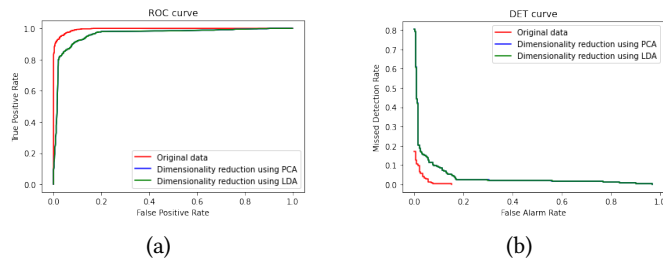


FIGURE 1.15

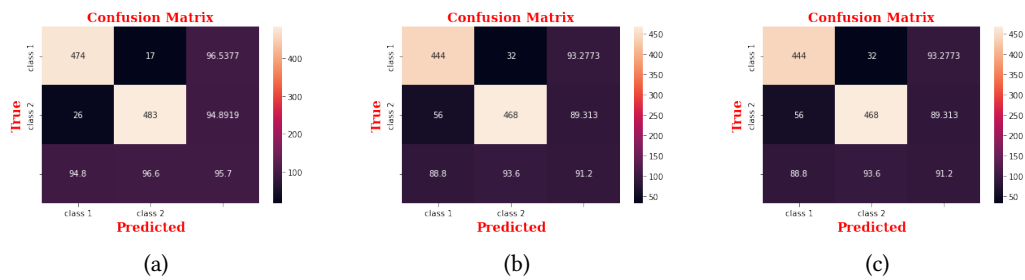


FIGURE 1.16

1.3.2 Image Dataset

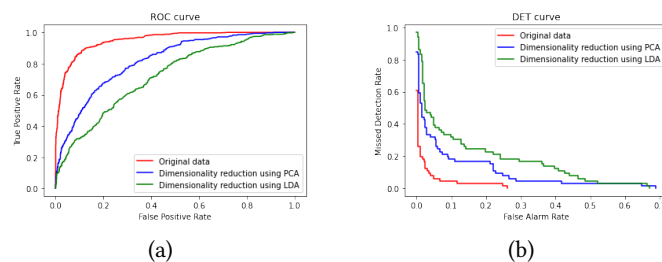


FIGURE 1.17

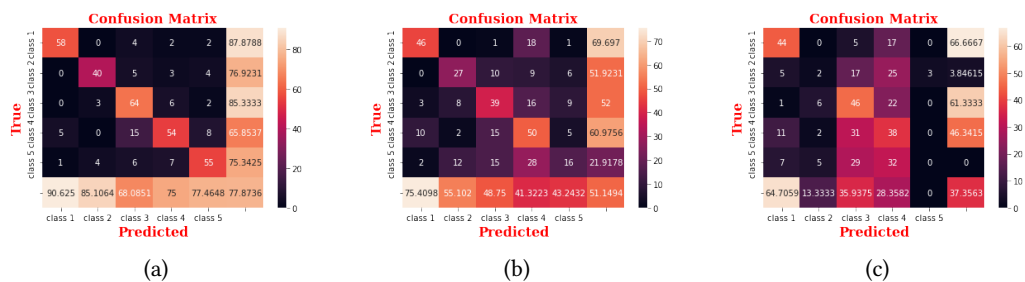


FIGURE 1.18

1.3.3 Isolated digits Dataset

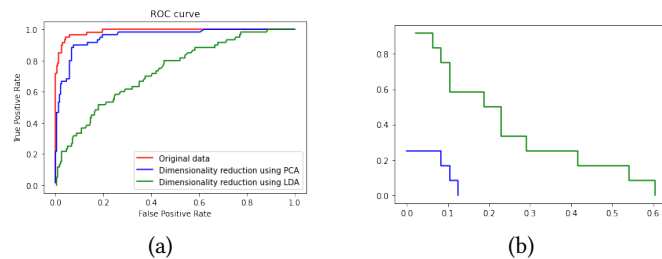


FIGURE 1.19

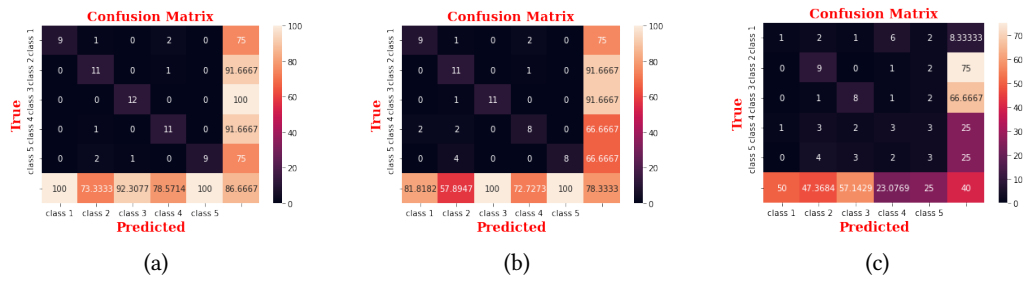


FIGURE 1.20

1.3.4 Handwritten character Dataset

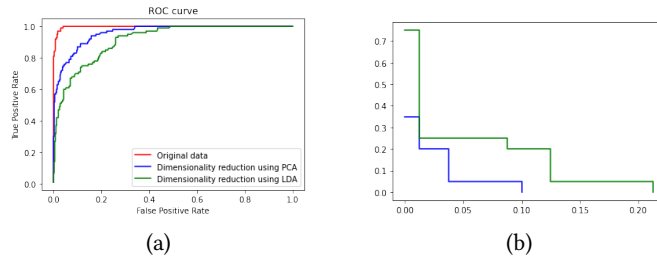


FIGURE 1.21

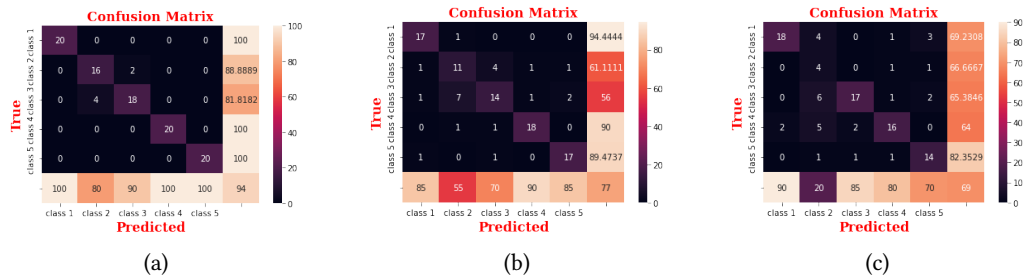


FIGURE 1.22

1.4 Artificial Neural Network

Results are arranged in the order of without dimension reduction, PCA, LDA. For PCA and LDA results are for fixed value of reduced dimension (NX4). N is largest data time-step. We used Scipy.signal Resample to make equal length time series.

1.4.1 Synthetic Dataset

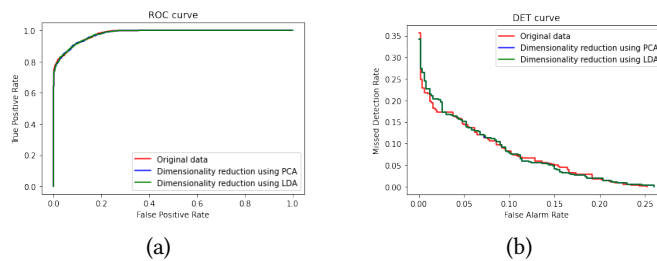


FIGURE 1.23

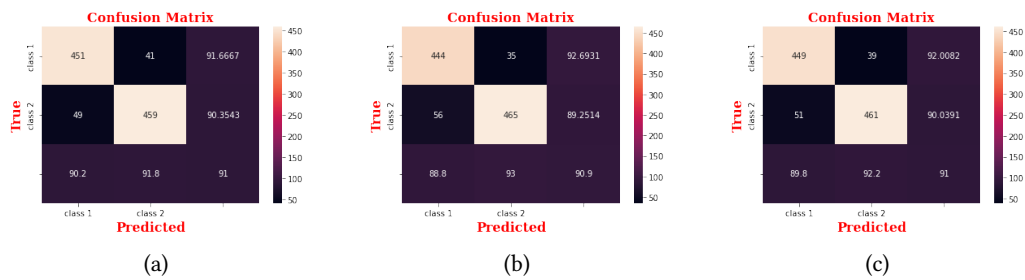


FIGURE 1.24

1.4.2 Image Dataset

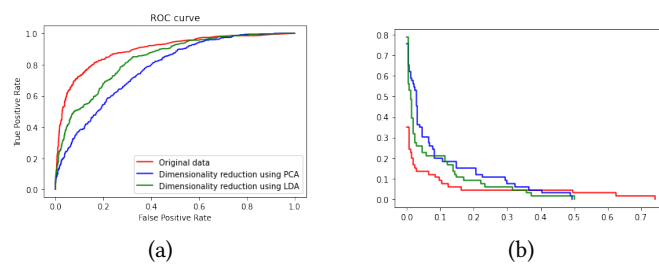


FIGURE 1.25

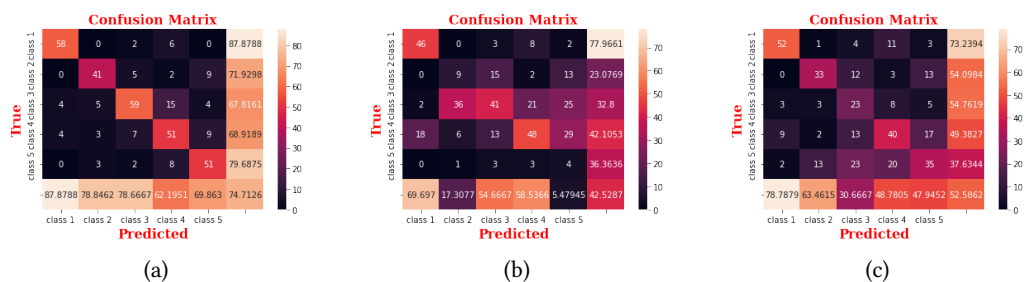


FIGURE 1.26

1.4.3 Isolated digits Dataset

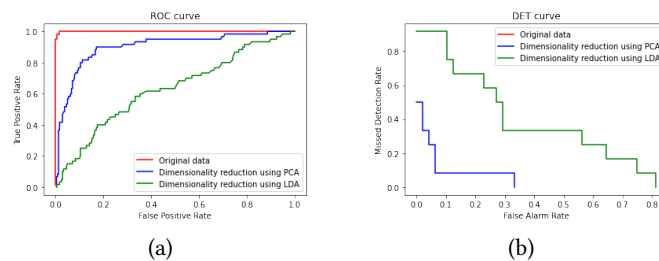


FIGURE 1.27

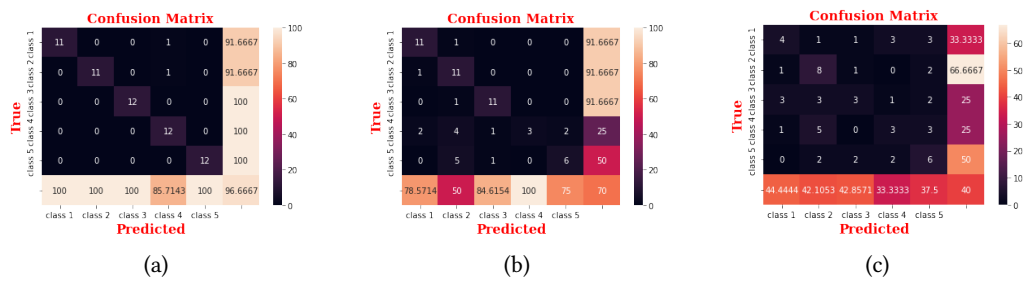


FIGURE 1.28

1.4.4 Handwritten character Dataset

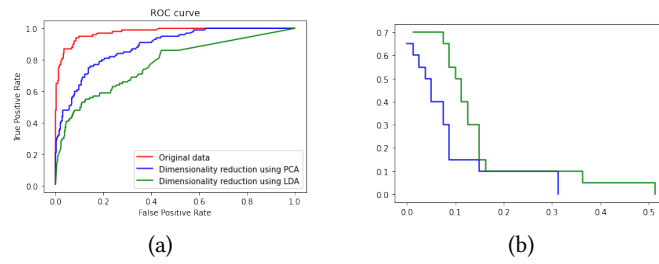


FIGURE 1.29

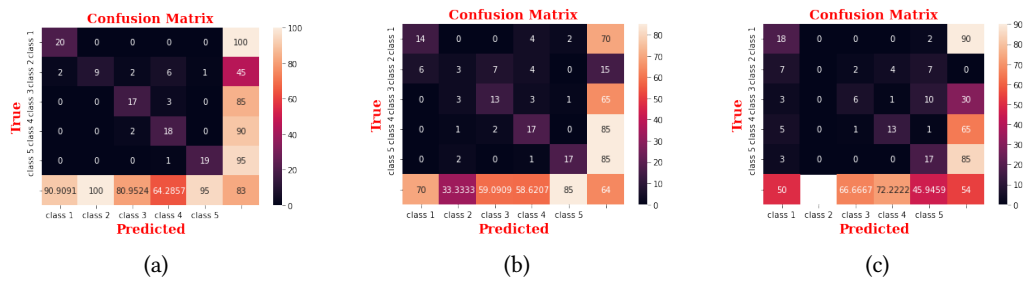


FIGURE 1.30